Republic of the Philippines

University of Cabuyao

(PAMANTASAN NG CABUYAO)

COLLEGE OF COMPUTING STUDIES

Katapatan Mutual Homes, Brgy. Banay-banay, City of Cabuyao, Laguna 4025

ITP101- QUANTITATIVE METHODS
MIDTERM LONG QUIZ

Student Name (LN, FN MI.):				**SCORE:**	
Section:		**Instructor:**	Ms. Sairine C. Pregonero	**Sem./ A.Y.**	2nd/25-26

Direction: Use the back of this paper and the provided answer sheet. Answer the following problems. Show your complete solution and all necessary computations. Include the formulation of hypotheses, computation of test statistics, decision rules, and conclusions where applicable. **Clearly indicate all formulas used and box your final answers for each computation.**

1.A bottled water company claims that the average volume of water in its bottles is 500 mL. A random sample of 36 bottles produced a mean volume of 495 mL. The population standard deviation is known to be 12 mL.
At the 5% level of significance, is there sufficient evidence to conclude that the average volume differs from 500 mL?

2. A school administrator claims that students spend an average of 5 hours per week using the library. A random sample of 25 students produced a mean of 4.2 hours with a sample standard deviation of 1.8 hours.
At the 5% level of significance, is there sufficient evidence to reject the administrator's claim?

3. A professor wants to know if there is a significant difference in test scores among three teaching methods.
Data

Method A	Method B	Method C
78	85	90
80	88	92
82	86	91

4. A researcher collected data on the number of hours students spend studying and their corresponding examination scores.

Study Hours (X)	Exam Score (Y)
2	65
3	70
4	75
5	80
6	85

Using the Pearson Product Moment Correlation Coefficient, is there a significant linear relationship between study hours and examination scores at the 5% level of significance?

5. A school principal wants to determine whether there is a significant relationship between students' IQ scores and Mathematics grades.

IQ Score (X)	Mathematics Grade (Y)
95	80
100	83
105	85
110	89
115	92

Using Pearson Correlation, is there sufficient evidence to conclude that a significant relationship exists between IQ scores and Mathematics grades at the 5% level of significance?

6. A retail company wants to predict monthly sales based on the number of advertisements released each month. Historical data were collected as follows:

Advertisements (X)	Monthly Sales (Y)
10	100
15	130
20	150
25	180
30	210

Using Linear Regression Analysis:
a. Determine the regression equation.
b. Predict the monthly sales when 50 advertisements are released.

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ANSWER SHEET

STATISTICAL CRITICAL VALUE TABLES

Reference tables for ITP101 Midterm Quiz | a = 0.05 (two-tailed)

Table 1 · Z Critical Values (Standard Normal Distribution)

Used for: Problem 1 (Z-Test). Reject H_0 if |Z_computed| > Z_critical.

Significance Level (α)	0.10	0.05	0.02	0.01
One-tailed Z_critical	1.282	**1.645**	2.054	2.326
Two-tailed Z_critical	1.645	**1.960**	2.326	2.576
Common use	Less strict	**Most common**	Strict	Very strict

Table 2 · t Critical Values (two-tailed)

Reject H_0 if |t_computed| > t_critical.

df	α=0.10	α=0.05	α=0.01		df	α=0.10	α=0.05 ★	α=0.01
1	6.314	12.706	63.657		**18**	1.734	2.101	2.878
2	2.920	4.303	9.925		**19**	1.729	2.093	2.861
3	2.353	3.182	5.841		**20**	1.725	2.086	2.845
4	2.132	2.776	4.604		**21**	1.721	2.080	2.831
5	2.015	2.571	4.032		**22**	1.717	2.074	2.819
6	1.943	2.447	3.707		**23**	1.714	2.069	2.807
7	1.895	2.365	3.499		**24**	1.711	2.064	2.797
8	1.860	2.306	3.355		**25**	1.708	2.060	2.787
9	1.833	2.262	3.250		**26**	1.706	2.056	2.779
10	1.812	2.228	3.169		**27**	1.703	2.052	2.771
11	1.796	2.201	3.106		**28**	1.701	2.048	2.763
12	1.782	2.179	3.055		**29**	1.699	2.045	2.756
13	1.771	2.160	3.012		**30**	1.697	2.042	2.750
14	1.761	2.145	2.977		**40**	1.684	2.021	2.704
15	1.753	2.131	2.947		**60**	1.671	2.000	2.660
16	1.746	2.120	2.921		**120**	1.658	1.980	2.617
17	1.740	2.110	2.898					

Table 3 · F Critical Values (α = 0.05 — upper tail)

Reject H_0 if F_computed > F_critical.

$df_2 \backslash df_1$	1	2	3	4	5	6	7	8	9	10
1	161.45	199.50	215.71	224.58	230.16	233.99	236.77	238.88	240.54	241.88
2	18.51	19.00	19.16	19.25	19.30	19.33	19.35	19.37	19.38	19.40
3	10.13	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81	8.79
4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.96
5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74
6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06
7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64
8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35
9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14
10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98
12	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80	2.75
15	4.54	3.68	3.29	3.06	2.90	2.79	2.71	2.64	2.59	2.54
20	4.35	3.49	3.10	2.87	2.71	2.60	2.51	2.45	2.39	2.35
24	4.26	3.40	3.01	2.78	2.62	2.51	2.42	2.36	2.30	2.25
30	4.17	3.32	2.92	2.69	2.53	2.42	2.33	2.27	2.21	2.16
40	4.08	3.23	2.84	2.61	2.45	2.34	2.25	2.18	2.12	2.08
60	4.00	3.15	2.76	2.53	2.37	2.25	2.17	2.10	2.04	1.99
120	3.92	3.07	2.68	2.45	2.29	2.18	2.09	2.02	1.96	1.91

Table 4 · Pearson r Critical Values (two-tailed)

Reject H_0 if |r_computed| > r_critical.

df	α=0.05	α=0.01		df	α=0.05	α=0.01
1	0.997	1.000		**14**	0.497	0.623
2	0.950	0.990		**15**	0.482	0.606
3	0.878	0.959		**16**	0.468	0.590
4	0.811	0.917		**17**	0.456	0.575
5	0.754	0.875		**18**	0.444	0.561
6	0.707	0.834		**19**	0.433	0.549
7	0.666	0.798		**20**	0.423	0.537
8	0.632	0.765		**25**	0.381	0.487
9	0.602	0.735		**30**	0.349	0.449
10	0.576	0.708		**40**	0.304	0.393
11	0.553	0.684		**60**	0.250	0.325
12	0.532	0.661		**100**	0.195	0.254
13	0.514	0.641				

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